

D883

Secondary Biology Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A student writes in a lab journal: giraffes grew long necks because they stretched to reach high leaves, and their babies inherited the longer necks. The journal is decorated with stickers and was turned in two days late. Which teacher response best corrects the idea?

- A. Giraffes evolved long necks because they needed them to survive.
- B. Giraffes already varied in neck length; those with longer necks survived and reproduced more, so the trait became common.
- C. Stretching does lengthen necks, but it takes many generations of stretching.
- D. Each giraffe evolves during its life to fit its environment.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

After years of heavy use of one pesticide, most beetles in a field are resistant. Which statement explains the result?

- A. The pesticide caused beetles to develop resistance so they could survive.
- B. Resistance appears in every population after exactly one generation.
- C. A few beetles were already resistant; they survived spraying and passed resistance on.
- D. Each beetle became resistant after being sprayed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A science fair project surveys a population of 1,000 flowers in a meadow near a state park. The meadow is mowed each fall by volunteers. White flowers (recessive, aa) make up 16% of the plants. Assuming Hardy-Weinberg equilibrium, what is the frequency of the recessive allele a?

- A. 0.6
- B. 0.16
- C. 0.48
- D. 0.4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

In a population at Hardy-Weinberg equilibrium, 9% of individuals show a recessive trait. What percentage are heterozygous carriers?

- A.** 49%
- B.** 91%
- C.** 21%
- D.** 42%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A case study describes two frog species that live in the same pond. The pond is inside a nature preserve with a paved walking trail. One species breeds in early March and the other in late April, so they never mate. What kind of barrier keeps them apart?

- A.** Geographic isolation
- B.** Postzygotic hybrid sterility
- C.** Behavioral isolation
- D.** Prezygotic temporal isolation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

In a kelp forest case study, sea otters are removed by overhunting. The kelp forest is a popular site for scuba tourism and has cold, clear water. Sea otters eat sea urchins, and sea urchins eat kelp. What is the most likely long-term result?

- A.** Sea urchins decrease and kelp grows thicker.
- B.** Kelp declines because tourists damage it.
- C.** Sea urchins increase and the kelp forest shrinks.
- D.** Sea urchins increase, and the kelp is unaffected.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Wolves are returned to a park where elk had overgrazed young willows along streams. What change should students predict for the willows?

- A.** Willows decline further as elk move into stream areas to escape wolves.
- B.** Willows decline because wolves eat willows.
- C.** Willows recover as elk numbers and grazing drop.
- D.** Nothing changes, because wolves only affect elk.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A deer herd on an island has a carrying capacity of 800 deer. The island has one ferry dock and a lighthouse built in 1890. Biologists record the population at 100, 400, and 780 deer in different years. At which size was the herd growing fastest?

- A.** 400 deer
- B.** 800 deer
- C.** 100 deer
- D.** 780 deer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A fish population is at its carrying capacity. Which statement is true?

- A.** The population is growing at its fastest rate.
- B.** The birth rate about equals the death rate, so the population stays near the same size.
- C.** The population will keep growing exponentially.
- D.** The birth rate is zero.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A wildfire burns a forest in a mountain valley, but the soil remains. Local news covered the fire for a week, and a scout troop later helped plant signs. Grasses and shrubs appear within a year. What process is this?

- A.** Primary succession
- B.** Secondary succession
- C.** Primary succession, because all plants were destroyed.
- D.** A climax community

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher shows a chart of a city's electricity coming from a coal plant built in the 1970s. The plant employs 300 people and sits on a river. When coal is burned, where does most of its stored carbon go?

- A.** Into the atmosphere as carbon dioxide
- B.** Into the ocean as dissolved oxygen
- C.** Into plants through photosynthesis
- D.** Into the ozone layer, making the ozone hole bigger

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A population of 20,000 prairie birds is reduced to 30 after a severe drought. The prairie is part of a restoration project funded by a regional nonprofit. Which outcome is most likely for the surviving population?

- A.** Reduced genetic diversity, because chance has a strong effect on allele frequencies in small populations.
- B.** Only helpful alleles will remain.
- C.** No change in allele frequencies, because drift only matters in large populations.
- D.** Higher genetic diversity, because survivors are the fittest.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A student writes in a lab journal: giraffes grew long necks because they stretched to reach high leaves, and their babies inherited the longer necks. ~~The journal is decorated with stickers and was turned in two days late.~~ Which teacher response best corrects the idea?

Target word: correct the misconception **Decides:** Traits acquired by use are not inherited; selection acts on existing variation.

A	Confusion set	Need does not produce heritable change.
B	Correct	Selection acts on heritable variation that already exists.
C	Inverted operation	Keeps the inheritance of acquired traits, just slows it down.
D	Adjacent category	Populations evolve over generations; individuals do not.

2. Answer C Concept 1 • transfer

After years of heavy use of one pesticide, most beetles in a field are resistant. Which statement explains the result?

Target word: explain resistance **Decides:** Resistant beetles existed before and survived.

A	Confusion set	Need does not cause the mutation.
B	Absolute language	Speed depends on variation and selection strength.
C	Correct	Selection by the pesticide favored existing resistant variants.
D	Adjacent category	Individuals do not evolve; populations do.

3. Answer D Concept 2

A science fair project surveys a population of 1,000 flowers in a meadow ~~near a state park. The meadow is mowed each fall by volunteers.~~ White flowers (recessive, aa) make up 16% of the plants. Assuming Hardy-Weinberg equilibrium, what is the frequency of the recessive allele a?

Target word: q **Decides:** $q^2 = 0.16$.

A	Inverted operation	That is p, the dominant allele frequency.
B	Dropped element	That is q^2 , not q.
C	Confusion set	That is 2pq, the carrier frequency.
D	Correct	$q = \text{square root of } 0.16 = 0.4$.

4. Answer D Concept 2 • transfer

In a population at Hardy-Weinberg equilibrium, 9% of individuals show a recessive trait. What percentage are heterozygous carriers?

Target word: 2pq **Decides:** $q^2 = 0.09$, so $q = 0.3$ and $p = 0.7$.

A	Confusion set	That is p^2 , the homozygous dominant frequency.
B	Inverted operation	That is everyone not showing the trait, not just carriers.
C	Dropped element	Forgets to multiply by 2.
D	Correct	$2 \times 0.7 \times 0.3 = 0.42$.

5. Answer D Concept 3

A case study describes two frog species that live in the same pond. ~~The pond is inside a nature preserve with a paved walking trail.~~ One species breeds in early March and the other in late April, so they never mate. What kind of barrier keeps them apart?

Target word: barrier type **Decides:** Different breeding times stop mating before a zygote forms.

A	Adjacent category	They share one pond, so location is not the barrier.
B	Inverted operation	No hybrids form, so the barrier acts before the zygote.
C	Confusion set	Behavioral isolation involves courtship signals, not timing.
D	Correct	Different breeding seasons prevent mating.

6. Answer C Concept 4

In a kelp forest case study, sea otters are removed by overhunting. ~~The kelp forest is a popular site for scuba tourism and has cold, clear water.~~ Sea otters eat sea urchins, and sea urchins eat kelp. What is the most likely long-term result?

Target word: effect two levels down **Decides:** Otter removal releases urchins, which overgraze kelp.

A	Inverted operation	Reverses the predator-prey effect.
B	Camouflage	The tourism detail does not decide the outcome.
C	Correct	A trophic cascade: fewer predators, more grazers, less kelp.
D	Dropped element	Stops one level short of the cascade.

7. Answer C Concept 4 • transfer

Wolves are returned to a park where elk had overgrazed young willows along streams. What change should students predict for the willows?

Target word: effect of restoring a predator **Decides:** Wolves reduce elk grazing pressure.

A	Inverted operation	Predation risk tends to reduce grazing near streams.
B	Confusion set	Wolves are carnivores; they affect willows through elk.
C	Correct	Restoring the predator reverses the cascade.
D	Absolute language	Effects pass down the food chain.

8. Answer A Concept 5

A deer herd on an island has a carrying capacity of 800 deer. ~~The island has one ferry dock and a lighthouse built in 1890.~~ Biologists record the population at 100, 400, and 780 deer in different years. At which size was the herd growing fastest?

Target word: size with fastest growth **Decides:** Logistic growth is fastest near $K/2$.

A	Correct	$K/2 = 400$, where the logistic growth rate peaks.
B	Wrong stem number	That is K itself, where growth stops.
C	Adjacent category	Per-capita rate is high, but total growth is lower than at $K/2$.
D	Inverted operation	Near K , growth slows almost to zero.

9. Answer B Concept 5 • transfer

A fish population is at its carrying capacity. Which statement is true?

Target word: condition at K **Decides:** At K, births and deaths balance.

A	Inverted operation	Fastest growth is near K/2.
B	Correct	Net growth is near zero at K.
C	Confusion set	Exponential growth has no K.
D	Absolute language	Births continue; they are balanced by deaths.

10. Answer B Concept 6

A wildfire burns a forest in a mountain valley, but the soil remains. ~~Local news covered the fire for a week, and a scout troop later helped plant signs.~~ Grasses and shrubs appear within a year. What process is this?

Target word: type of succession **Decides:** Soil remained after the disturbance.

A	Confusion set	Primary begins on bare rock with no soil.
B	Correct	Recovery starts from existing soil.
C	Dropped element	Ignores that the soil remained, which is what decides it.
D	Adjacent category	That is the end stage, not the process of regrowth.

11. Answer A Concept 7

A teacher shows a chart of a city's electricity coming from a coal plant built in the 1970s. ~~The plant employs 300 people and sits on a river.~~ When coal is burned, where does most of its stored carbon go?

Target word: direction of carbon flow **Decides:** Combustion releases CO₂ into the air.

A	Correct	Burning moves carbon from long-term storage to the air.
B	Confusion set	Mixes up carbon and oxygen.
C	Inverted operation	Photosynthesis removes CO ₂ from the air; burning adds it.
D	Adjacent category	The ozone hole is a separate issue from CO ₂ and the greenhouse effect.

12. Answer A Concept 8

A population of 20,000 prairie birds is reduced to 30 after a severe drought. ~~The prairie is part of a restoration project funded by a regional nonprofit.~~ Which outcome is most likely for the surviving population?

Target word: effect of a bottleneck **Decides:** Small population size makes genetic drift strong.

A	Correct	A bottleneck makes drift strong and lowers diversity.
B	Confusion set	Drift is random; it can remove helpful alleles.
C	Inverted operation	Drift is strongest in small populations.
D	Inverted operation	Bottlenecks lower diversity.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Natural selection acts on existing variation	Individuals with heritable traits that help them survive and reproduce leave more offspring, so the population changes over generations; the variation must already exist.	Thinking organisms change because they need to, or that individuals evolve during their lifetime.	Wording like needed, tried, or adapted within one lifetime.
2	Hardy-Weinberg frequencies	In a population at equilibrium, $p + q = 1$ and $p^2 + 2pq + q^2 = 1$; the recessive phenotype frequency equals q^2 .	Setting q equal to the recessive phenotype frequency instead of its square root.	A percentage of individuals showing a recessive trait and a question about carriers or allele frequency.
3	Reproductive isolation	Prezygotic barriers stop mating or fertilization; postzygotic barriers act after a hybrid zygote forms, such as sterile or weak hybrids.	Classifying sterile hybrids as prezygotic, or treating any difference as a species barrier.	Whether a zygote forms: timing, behavior, or habitat vs. hybrid sterility.
4	Keystone species and trophic cascades	Removing a keystone predator changes the whole community through a chain of effects down the food web, often out of proportion to its numbers.	Thinking the effect stops at the prey species, or that the most common species is the keystone.	A predator removed or restored, with effects asked two levels down.
5	Logistic growth and carrying capacity	Population growth slows as it nears carrying capacity K ; growth is fastest at about $K/2$ and near zero at K .	Thinking growth is fastest when the population is largest.	A population size compared with K and asked when growth is fastest or slowest.
6	Primary vs. secondary succession	Primary succession starts on bare rock with no soil and pioneer species such as lichens; secondary succession follows a disturbance that leaves soil in place.	Calling any post-fire regrowth primary succession.	Whether soil remains after the disturbance.
7	The carbon cycle and human change	Burning fossil fuels moves carbon from long-term storage into the atmosphere as CO_2 ; photosynthesis removes CO_2 ; the added CO_2 strengthens the greenhouse effect.	Mixing up the ozone hole with the greenhouse effect, or thinking respiration and photosynthesis only affect oxygen.	A carbon flow direction question: into or out of the atmosphere.
8	Genetic drift in small populations	Random changes in allele frequency are strongest in small populations and can remove alleles regardless of fitness; bottlenecks reduce genetic diversity.	Thinking drift is a kind of natural selection, or that it matters most in large populations.	A population that crashed to a few individuals, or a chance event.

Course scope (public description): The Secondary Biology course offers a comprehensive understanding of biological principles and processes, starting from the fundamental cellular structures and functions to the complexities of ecosystem dynamics. Students will delve into key topics such as genetics, evolution, and ecological interactions, gaining insights into the diversity of life and the interconnectedness of biological systems. The course emphasizes critical thinking and problem-solving skills, enabling learners to analyze biological phenomena and

apply their knowledge to real-world challenges. Through a blend of theoretical knowledge and practical application, students will engage in hands-on activities, scenarios, and case studies to deepen their understanding. Additionally, the course highlights the impact of environmental changes on ecosystems, fostering an appreciation for conservation and sustainability. By the end of the course, learners will be well-prepared to address various biological challenges and pursue further studies or careers in the field of biology education.